

22) Show that if A and B are sets with $A \subseteq B$, then $A \cup B = B$, i.e. $x \in A$

a) $A \cup B = B$.

Solution: Given, $A \subseteq B$. Let, x be an arbitrary element of A . By Defn. 3 from Section 2.1, $x \in B$.

Let, y be an arbitrary element of $A \cup B$, i.e. $y \in A \cup B$. By Defn. 1, $y \in A$ or $y \in B$. By defn. of OR, at least one of $y \in A$ or $y \in B$ is True.

Let, $y \in A$ be True. By Defn. 3 from Section 2.1, $y \in B$. If $y \in B$ is True we are done.

By Defn. 3 from Section 2.1, $A \cup B \subseteq B$

Let, x be an arbitrary element of B , i.e. $x \in B$. By Defn. of OR, $x \in A \vee x \in B$. By Defn. 1, $x \in A \cup B$.

By Defn. 3 from Section 2.1, $B \subseteq A \cup B$.

$\therefore A \cup B = B$ $\square$

b) $A \cap B = A$

Solution: Let, x be an arbitrary element of $A \cap B$, i.e. $x \in A \cap B$. By Defn. of AND, $x \in A$.

By Defn. 2, $x \in A \wedge x \in B$. By Defn. 3 from Section 2.1, $A \cap B \subseteq A$

Let x be an arbitrary element of A , i.e. $x \in A$. By Defn. 3 from Sec-2.1, $\because A \subseteq B \therefore x \in B$.

By Defn. of AND, $x \in A \wedge x \in B$. By Defn. 2, $x \in (A \cap B)$.

By Defn. 3 from Section 2.1, $A \subseteq A \cap B$

$\therefore A \cap B = A$ $\square$

23) Prove the first associative law from Table 1 by showing that if A, B and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap C$

Solution: $A \cup (B \cap C) = \{x | x \in A \vee (x \in B \cap C)\}$ (Defn. 1)
 $= \{x | x \in A \vee (x \in B \wedge x \in C)\}$ (Defn. 1)
 $= \{x | (x \in A \vee x \in B) \wedge x \in C\}$ (1st Associative Law of Proposition)
 $= \{x | (x \in A \cup B) \wedge x \in C\}$ (Defn. 1) $= (A \cup B) \cap C$ (Defn. 1) $\square$